Better Slots, Better Worlds: Representation Quality & Robustness in Object-Centric World Models

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Abstract

Learning world models from offline trajectories enables agents to accomplish different tasks through planning. Object-centric (OC) representations, which decompose a scene into a set of slots that bind to its objects, have been proposed as an inductive bias for world models that are more sample-efficient and generalize better. Yet prior object-centric world models (OCWMs) take the slot encoder as given and evaluate only in-distribution, leaving open whether the object-centric bias actually delivers for *planning* and what within the OCWM drives it. We conduct a controlled study of OCWMs for visual model-predictive control along two axes: object-centric representation quality and generalization under distribution shift relative to scene-centric models. We find that (i) planning success correlates positively with unsupervised slot-quality metrics (FG-ARI, mBO), though the gains saturate at high slot quality; (ii) with well-bound slots, the auxiliary proprioception inputs and masking inductive bias that prior methods relied on become unnecessary; and (iii) an OCWM with well-bound slots plans more robustly under unseen distribution shifts than the end-to-end trained scene-centric LeWM, while DINO-WM, built on similar frozen pretrained features, remains comparably robust — suggesting that pretrained visual representations are an important contributor to robustness.

1 Introduction

A central promise of world models (Ha & Schmidhuber, 2018) is that an agent can learn its environment's dynamics from offline trajectories, and then plan toward arbitrary goals at test time (Zhou et al., 2025; Maes et al., 2026b). Since offline action-conditioned trajectories are costly to collect, the representation a world model uses matters for sample efficiency and generalization. Prior work spans a spectrum of representations: scene-centric models predict over DINOv2 patch features (Oquab et al., 2024) (DINO-WM) or a single global latent (LeWM), while object-centric models predict over a set of object slots (C-JEPA) (Nam et al., 2026). Object-centric representations are appealing in principle: by factorizing a scene to match its compositional, causal structure (Locatello et al., 2020; Schölkopf et al., 2021), they should allow a world model to generalize under distribution shifts. Two assumptions behind this appeal, however, are untested. First, the slot encoder is trained separately from the world model and selected using unsupervised slot-quality metrics (FG-ARI (Greff et al., 2019) and mBO (Pont-Tuset et al., 2017)) that reward clean object masks rather than planning success; whether better-scoring slots yield better downstream planning is therefore unknown. Second, whether the promised generalization holds for planning under distribution shift is untested.

We address both with a controlled study of OCWMs for visual model-predictive control on 2D PushT and 3D OGBench-Cube, along two axes: for quality, we train world models on SlotContrast (Manasyan et al., 2025) checkpoints of varying slot quality and measure planning success; for generalization, we compare our OCWM against scene-centric world models (DINO-WM, LeWM) under matched conditions. Our findings are:

- **Quality.** Planning success correlates positively with unsupervised slot-quality metrics (FG-ARI, mBO), with gains saturating at high quality; with well-bound slots, the OCWM no longer needs the auxiliary proprioception inputs and masking inductive bias that prior methods relied on.
- **Robustness.** Under unseen distribution shifts, the OCWM degrades the least, while DINO-WM remains similarly robust and LeWM degrades substantially; together, these results suggest that planning over frozen pretrained visual representations is associated with improved robustness.

2 Related Work

OCWMs such as OC-STORM and C-JEPA (Zhang et al., 2025; Nam et al., 2026; Spieler et al., 2026) are evaluated only in-distribution. While Wang et al. (2026) evaluate their Dyn-O OCWM on held-out Procgen levels, they report only rollout visual fidelity rather than control metrics (returns or planning performance). Outside of world modeling, Dittadi et al. (2022) provide a controlled study of OOD generalization for slot representations on downstream property prediction, and Yoon et al. (2023) examine pre-trained OC representations for model-free RL. Our study fills this gap for OCWMs by sweeping slot quality with a fixed dynamics model and evaluating per-object visual, frame-level, and dynamics shifts against scene-centric WMs under matched conditions. Related works on object-centric learning and world modeling are summarized in Appendix D.

3 Experiments and Results

We investigate three research questions: (1) Does slot quality impact planning success? (2) Do better slots allow sidestepping redundant auxiliary inputs and additional training objectives? (3) Do object-centric representations plan more robustly under distribution shifts than scene-centric ones?

3.1 Setup

Our setup builds on C-JEPA (Nam et al., 2026), an object-centric world model with a non-causal variant (OC-JEPA) and a causal one (C-JEPA); both encode each frame into object slots with VideoSAUR (Zadaianchuk et al., 2023) and predict future slots with a transformer dynamics module. C-JEPA augments OC-JEPA with a causal inductive bias: at each step, it partitions the slots into masked and unmasked subsets and predicts the masked slots from the unmasked ones, encouraging the model to capture *inter-object interactions* rather than relying on self-dynamics.

We adopt this framework for visual model-predictive control with several improvements: we replace VideoSAUR with SlotContrast (Manasyan et al., 2025), which provides stronger temporal consistency and eliminates the need for Hungarian matching across slots and we update DINOv2 with DINOv3 as the feature extractor. Unless noted otherwise, our default world model, **SlotContrast-WM**, is the non-causal OC-JEPA backbone with neither the slot-masking objective nor the auxiliary proprioception token. Extended details are in Appendix A.2.

Environments and Baselines We evaluate our approach on the 2D PushT and 3D OGBench-Cube environments. We compare SlotContrast-WM against two scene-centric baselines: DINO-WM, which utilizes frozen DINOv2 patch tokens, and LeWM, which relies on the global CLS token of an end-to-end trained ViT (Dosovitskiy et al., 2021). Extended hyperparameters and planning configurations are in Appendix A.3.

Metrics We assess slot quality with two unsupervised metrics: the Foreground Adjusted Rand Index (*FG-ARI*) (Greff et al., 2019), measuring how well objects are separated into slots, and mean Best

Overlap (*mBO*) (Pont-Tuset et al., 2017), an IoU-based segmentation score that assesses mask sharpness. We use their *video* variants because, for world modeling, temporally consistent slot identity matters more than per-frame segmentation quality. For planning, we report task success rate, with criteria detailed in Appendix A.1.

3.2 Does slot quality impact planning success?

C-JEPA, the OCWM we build on, encodes frames with a VideoSAUR encoder whose slots are poorly bound, fragmenting objects and bleeding them into the background (Figure 1, right), yet C-JEPA reports strong planning performance with it. If the object-centric inductive bias is what aids planning, a representation that so visibly fails to isolate objects ought to plan poorly, raising a more basic question: *does unsupervised slot quality bear on planning success at all*? To test this, we hold the dynamics model and planner fixed and train SlotContrast-WM on intermediate SlotContrast checkpoints spanning a range of slot quality, measuring downstream planning success.

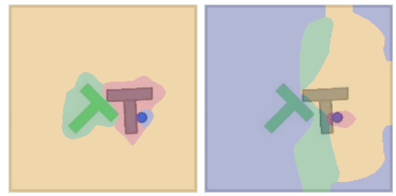

Figure 1: Slot decomposition on PushT. **Left:** SlotContrast binds each object (agent, T-block, goal) to a dedicated slot. **Right:** VideoSAUR fragments objects across slots and mixes them with the background.

As shown in Figure 2, downstream planning performance closely tracks the emergence of object separation. On OGBench-Cube, however, the same sweep saturates early (Appendix B.1, Figure 5).

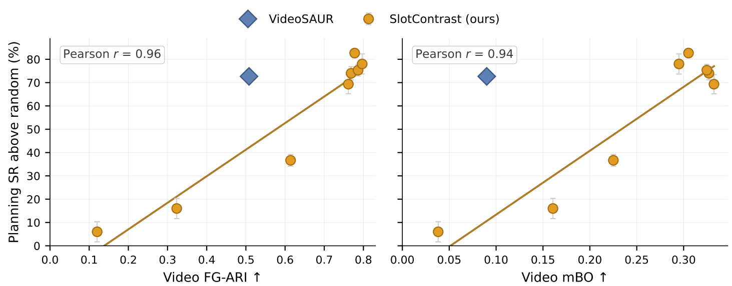

Figure 2: Across SlotContrast checkpoints (orange), planning success rate above the random-policy baseline (2%) is positively correlated with video slot-quality metrics: FG-ARI (left, Pearson $r = 0.96$) and mBO (right, $r = 0.94$). VideoSAUR (blue diamond) uses the same minimal configuration (no slot masking or proprioception).

Takeaway 1: *Planning success is positively correlated with slot quality with no change to the dynamics model, but the gains saturate: at high slot quality, where the metrics themselves plateau, better-bound slots yield little additional planning success.*

3.3 Do better slots sidestep auxiliary inputs and training objectives?

Why does VideoSAUR plan well under C-JEPA despite its poorly bound slots (Figure 1)? We hypothesize that the auxiliary mechanisms of *masked-slot history prediction* objective and the *auxiliary proprioception* token, which C-JEPA inherits, do not add predictive power but compensate for weak slots. We test this by sweeping `num_masked_slots` $\in \{0, 1, 2\}$, the number of slots masked at each training step and predicted from the rest, with and without proprioception on a high-quality (SlotContrast) and a low-quality (VideoSAUR) encoder (further visualizations in Figure 4a).

Without proprioception or masking, SlotContrast-WM reaches 84.7% (± 1.9) SR, and adding the proprioception token barely helps ($+0.6$ pp). This minimal configuration already matches, within seed variance, the 85.3% (± 3.4) of the full C-JEPA recipe (VideoSAUR with proprioception *and* masking) and exceeds VideoSAUR (74.7%) by 10 pp. Without proprioception, masking monotonically degrades both encoders (Appendix B, Table 3, $-$prop rows), and the same ordering holds on OGBench-Cube. Masking helps only when paired with proprioception on the weak encoder

(73.3 → 85.3%), suggesting it leans on a proprioceptive shortcut rather than exploiting visual object interactions.

Takeaway 2: *In our manipulation tasks, a sufficiently object-centric representation makes both slot history masking and proprioception unnecessary: they only compensate for weak representations.*

3.4 Do object-centric representations allow robustness to distribution shifts?

We evaluate world-model robustness under visual and dynamics shifts (variations visualized in Appendix C). Under object-level appearance shifts, SlotContrast-WM retains the highest success rates, and DINO-WM (patch features) degrades only moderately, whereas LeWM (CLS token of an end-to-end trained ViT) collapses (Figure 3). DINO-WM plans over frozen pretrained DINO features directly, while SlotContrast-WM plans over object slots that Slot Attention extracts from the same features; the shared pretrained foundation appears to be the main source of robustness to appearance shifts. Frame-level perturbations such as a changed background color are difficult for all models and most damaging for LeWM, whose global CLS token they fundamentally alter. On OGBench-Cube, SlotContrast-WM and DINO-WM stay close to their in-distribution success across all shifts, while LeWM degrades under scene-level shifts (Appendix C.4, Figure 8). Notably, every model fails under geometric variations, since changing an object's shape alters its contact dynamics.

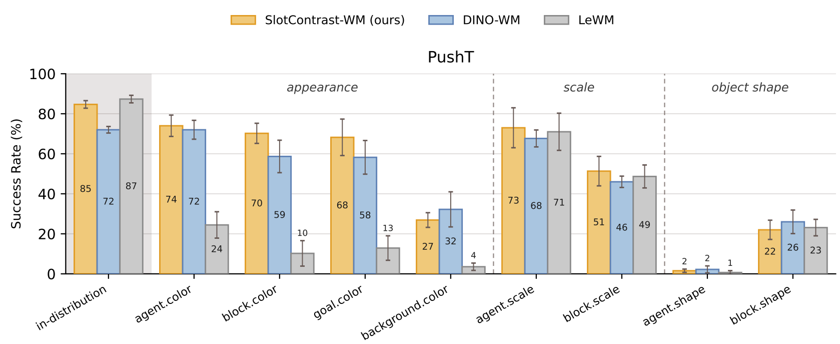

Figure 3: Planning success under distribution shift on PushT. OCWM (SlotContrast-WM) versus scene-centric baselines (DINO-WM, LeWM).

Takeaway 3: *World models planning over frozen pretrained features — object-centric or not — tolerate appearance and frame-level shifts far better than the end-to-end trained LeWM.*

4 Conclusion

Our controlled study isolates what makes object-centric world models work for planning: representation quality is the dominant factor — planning success closely tracks slot quality and saturates once slots are well-bound, at which point the auxiliary proprioception input and slot-masking objective become unnecessary. Given good slots, the resulting OCWM is also the most robust model in our study, though DINO-WM, built on similar frozen pretrained features, retains much of this robustness while the end-to-end trained LeWM degrades severely. In short, better slots enable stronger object-centric planning, while frozen pretrained visual representations appear to be an important ingredient for robustness under distribution shift.

Limitations & future work. Unsupervised slot metrics are less informative when task-relevant objects are small relative to the scene (e.g., OGBench-Cube), motivating task-aware quality measures and end-to-end encoder–world-model training. Future work should test these findings in more diverse environments—more objects, varied scales, richer dynamics—to probe OCWM's robustness and compositional generalization.

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A Experiment Details

A.1 Environments and Datasets

We evaluate our framework across two distinct physical manipulation domains.

PushT A 2D planar manipulation task where a circular agent must push a T-shaped block to align with a specific target pose.

- **Dataset:** The offline dataset consists of 18,500 expert demonstrations augmented with varying levels of action noise. This dataset is adopted from Maes et al. (2026b), utilizing the original generation protocol introduced by Zhou et al. (2025).
- **Success Criterion:** For goal-conditioned planning, an episode is marked successful if the combined spatial translation error of both the agent and the block is less than 20 pixels, and the block's angular orientation error is less than $\pi/9$ radians ($\approx 20°$) relative to the goal configuration. This follows the evaluation protocols established in prior work (Zhou et al., 2025; Maes et al., 2026b).

OGBench-Cube A 3D object manipulation environment introduced by Park et al. (2025). The workspace features a simulated UR5e robotic arm equipped with a Robotiq 2F-85 adaptive gripper, tasked with interacting with a colored cube to achieve a spatial goal configuration.

- **Dataset:** The offline dataset contains 10,000 heuristic-generated demonstrations (200 transitions per episode), adopted from Maes et al. (2026b).
- **Success Criterion:** An episode is considered successful if the Euclidean distance between the cube's final position and its goal target position is less than 4 cm. In contrast to the PushT, this environment does not enforce kinematic constraints on the final pose of the agent (robotic end-effector), making the random policy achieve high planning success rate. This setup follows the baseline methods and is a known limitation of the environment evaluation.

Both environments are implemented in `stable-worldmodel` (Maes et al., 2026a) and we utilize this framework for all evaluations throughout this paper.

A.2 Object-Centric Encoders

SlotContrast. SlotContrast (Manasyan et al., 2025) is a video object-centric learning framework designed to discover temporally consistent object representations from videos. The model comprises three main components: (1) a pre-trained self-supervised dense feature encoder, such as DINOv2 (Oquab et al., 2024), which extracts rich spatial features from each frame; (2) a Recurrent Slot Attention (Locatello et al., 2020; Kipf et al., 2022) module that groups these features into object-centric slots while modeling their temporal evolution across frames; and (3) a decoder that reconstructs the dense self-supervised features from the slot representations. Training is guided by two complementary objectives. A feature reconstruction loss encourages the slots to capture the information necessary to reconstruct the encoder features, while a temporal consistency loss contrasts slot representations across consecutive frames. This temporal contrastive objective promotes stable slot assignments and enables the model to learn object representations that remain consistent over time, leading to improved object discovery and tracking in dynamic video scenes.

Our setup. We adopt SlotContrast as our object-centric backbone, replacing the original DINOv2 encoder with the more recent DINOv3 (Siméoni et al., 2025), which provides stronger dense features. We otherwise follow the original training objectives. The full list of hyperparameters for PushT and OGBench-Cube is given in Table 1.

VideoSAUR. VideoSAUR (Zadaianchuk et al., 2023) is another object-centric video representation learning framework that combines DINO-based semantic features with Slot Attention for object discovery. Its key contribution is a temporal feature similarity objective that exploits motion bias

Table 1: Hyperparameters of SlotContrast Model on PushT and OGBench-Cube.

Hyperparameter	**PushT**	**OGBench**
Training Steps	100k	100k
Batch Size	128	128
Training Segment Length	4	4
Learning Rate Warmup Steps	2500	2500
Optimizer	Adam	Adam
Peak Learning Rate	0.0004	0.0004
ViT Architecture	DINOv3 Small	DINOv3 Small
Initialization	FixedLearnedInit	FixedLearnedInit
Patch Size	16	16
Feature Dimension (D_{feat})	384	384
Gradient Norm Clipping	0.05	0.05
Image Specifications		
Image / Crop Size	256	256
Cropping Strategy	Full	Full
Augmentations	Rand. Horizontal Flip	Rand. Horizontal Flip
Image Tokens	256	256
Slot Attention		
Slots	4	3
Iterations (first / other frames)	3 / 2	3 / 2
Slot Dimension (D_{slots})	128	128
Predictor		
Type	Transformer	Transformer
Layers	1	1
Heads	4	4
Decoder		
Type	MLP	MLP
Loss Parameters		
Softmax Temperature (τ)	0.1	0.01
Slot–Slot Contrast Weight (α)	0.01	0.001

to promote object disentanglement by preserving semantic and temporal relationships among DINO patch embeddings. This objective can be used either alongside a reconstruction loss or as a standalone training signal. However, because VideoSAUR does not explicitly constrain slot identities to remain temporally consistent, slot correspondences across frames may need to be recovered post hoc using Hungarian matching or similar methods.

Qualitative Mask Analysis. To visually validate the differences in slot quality between the evaluated encoders, we extract and render the learned slot masks over an episode. As shown in Figure 4a, SlotContrast successfully isolates the agent, the T-block, and the background into distinct, temporally consistent slots in the PushT environment. In contrast, VideoSAUR suffers from severe feature entanglement, partitioning the space into Voronoi-like regions where object identities bleed across slot boundaries (a known limitation of Slot Attention-based methods (Sajjadi et al., 2022)). Figure 4b further validates SlotContrast's object-centricity in 3D settings, demonstrating clean isolation and tracking of the robotic arm, the gripper, and the cube, even in contact events when the end-effector grasps the cube.

A.3 Visual Model-Based Planning

All world models are evaluated under the same Cross-Entropy Method (CEM) (Rubinstein, 1999) planner budget and identical context history.

Planner Cost Functions The CEM planner optimizes action sequences by minimizing the distance between the model's predicted latent state $\hat{Z}$ and the target goal state Z_{goal}. While all models utilize an L_2 norm formulation, the structure of the representation dictates exactly how this cost is computed:

- **LeWM:** Evaluates the distance between the single, global `[CLS]` vectors:

$$J = ||\hat{z}^{\text{cls}} - z^{\text{cls}}_{\text{goal}}||_2^2$$

- **DINO-WM:** Evaluates the mean squared error across the rigid spatial grid of P patches:

$$J = \frac{1}{P}\sum_{p=1}^{P} ||\hat{z}_p - z_{p,\text{goal}}||_2^2$$

- **SlotContrast-WM (Ours):** Because SlotContrast enforces strict temporal consistency, slot identities are preserved across time. This allows us to compute a direct, one-to-one distance across the K slots without any heuristic matching:

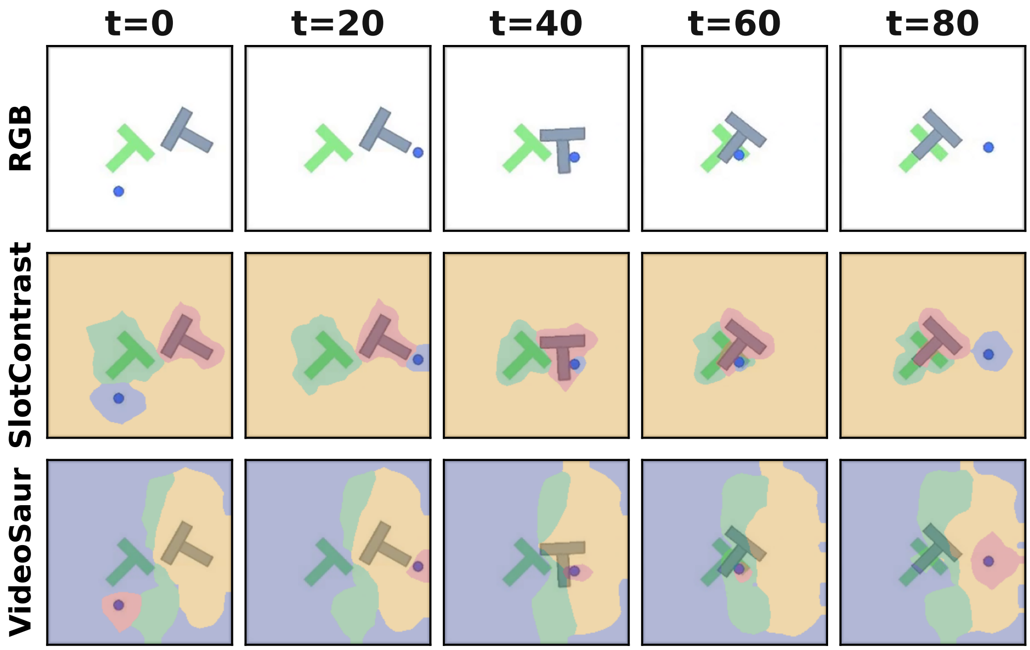
$$J = \frac{1}{K}\sum_{k=1}^{K} ||\hat{z}_k - z_{k,\text{goal}}||_2^2$$

- **C-JEPA (VideoSAUR):** VideoSAUR does not guarantee temporal consistency. Therefore, evaluating the cost requires solving a bipartite matching problem via the Hungarian algorithm at every single planning step to align predicted slots to goal slots, where Π is the set of all possible permutations:

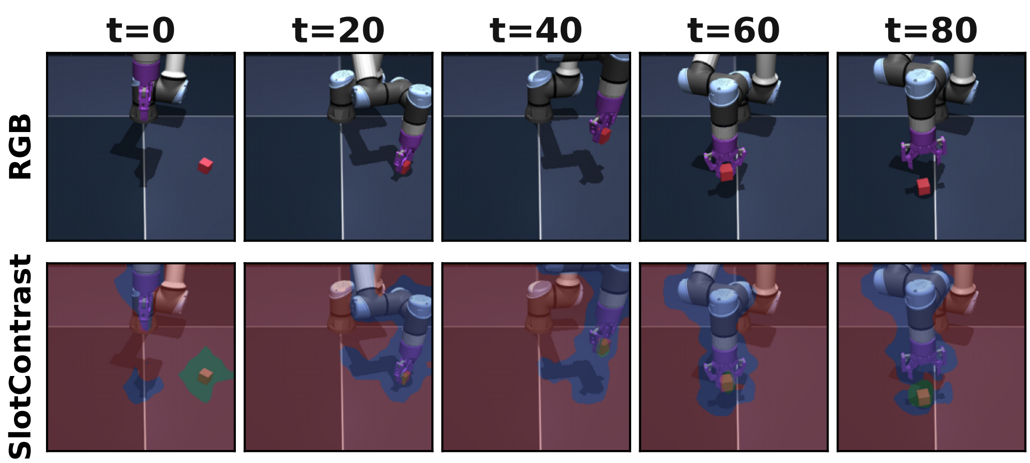
$$J = \min_{\pi \in \Pi} \frac{1}{K}\sum_{k=1}^{K} ||\hat{z}_k - z_{\pi(k),\text{goal}}||_2^2$$

Where specified, this visual cost is linearly combined with an auxiliary proprioceptive cost, scaled by `proprio_cost_weight` ($W \in \{0, 1\}$), detailed further in Appendix B.2.

All reported planning results are averaged over three random seeds. The fixed hyperparameters used for the CEM optimization per environment are detailed in Table 2.



(a) PushT Slot Masks: SlotContrast accurately isolates the physical objects (agent, T-block) from the background. VideoSAUR produces entangled partitions where object features bleed across slots.



(b) OGBench-Cube Slot Masks: SlotContrast successfully discovers and tracks 3D components (robotic arm and its shadow, cube, background) with temporal consistency over the episode. Notably, robotic arm and its shadow are assigned to the same slot.

Figure 4: Qualitative comparison of learned object-centric representations over 80 timesteps.

Table 2: Cross-Entropy Method (CEM) planning parameters used across all evaluated world models.

Parameter	**PushT**	**OGBench-Cube**
Environment Steps Horizon (H)	25	25
Number of evaluated goals	50	50
Frameskip	5	5
Latent Planning Horizon (H, frameskipped)	5	5
CEM Samples (N_{samples})	300	300
CEM Iterations (N_{iter})	30	30
Top Elites (K)	30	30
Initial Sampling Variance	1.0	1.0
MPC Execution Steps (frameskipped)	5	5

B Analyses on Encoder Quality

B.1 Slot Quality vs. Planning on OGBench-Cube

Figure 5 repeats the slot-quality sweep of the main text (Figure 2) on OGBench-Cube. Unlike PushT, the metrics start high and the relationship saturates early: FG-ARI and mBO are aggregated over the whole scene, where the large, easily segmented robot arm dominates the score while the small, task-relevant cube contributes little.

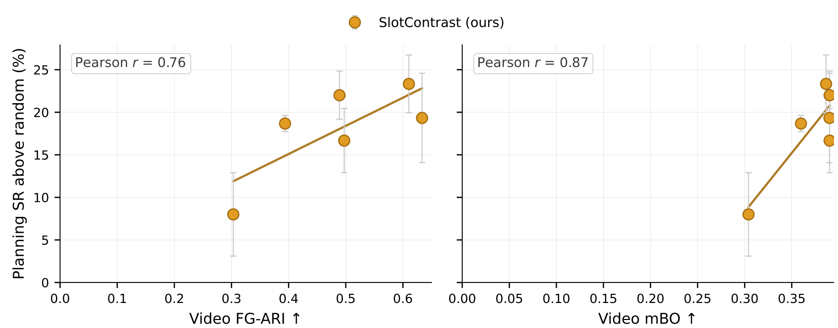


Figure 5: Slot quality vs. planning success on OGBench-Cube. Success rate is reported relative to the random-policy baseline (48%)

B.2 Role of Slot Masking and Proprioception Token

We sweep `num_masked_slots` $\in \{0, 1, 2\}$ crossed with `use_proprio` $\in \{\text{true, false}\}$, separately for the two encoder families. Each cell trains a C-JEPA WM with respective object-centric encoder(SlotContrast, VideoSAUR) on the same expert dataset with the same hyperparameters.

The same ordering holds on OGBench-Cube. With the SlotContrast-WM and no proprioception, masking provides no clear benefit in planning performance: nms=0 reaches $72.7 \pm 3.4\%$ SR versus $70.0 \pm 2.8\%$ for nms=1.

Table 3: Planning SR (%) on PushT evaluating `num_masked_slots` and proprioception across strong (SlotContrast) and weak (VideoSAUR) encoders. Without proprioception (−prop), masking strictly degrades planning for both encoders.

Encoder	Proprio	nms=0	nms=1	nms=2
SlotContrast	+prop	**85.3** $\pm$3.4	82.7 $\pm$1.9	82.7 $\pm$3.4
SlotContrast	−prop	**84.7** $\pm$1.9	72.0 $\pm$3.3	34.0 $\pm$6.5
VideoSAUR	+prop	73.3 $\pm$5.7	**85.3** $\pm$3.4	83.3 $\pm$4.1
VideoSAUR	−prop	**74.7** $\pm$1.9	52.0 $\pm$5.9	49.3 $\pm$6.6

C More Visualizations and Robustness Details

C.1 OOD Variation Groups

Each world model is trained on a single in-distribution rendering and evaluated on a fixed grid of unseen variations, changing one factor at a time. We use the same grouping as in the main-text figures. PushT variations are grouped into *appearance* (object and background colour), *scale* (object size), and *object shape*. OGBench-Cube variations are grouped into *object-level* (cube and agent) and *scene-level* (floor, camera, and lighting). The exact factors and values are listed in Tables 4 and 5.

C.2 PushT Variation Suite

Figure 6 shows one representative variation per factor; Table 4 lists all values.

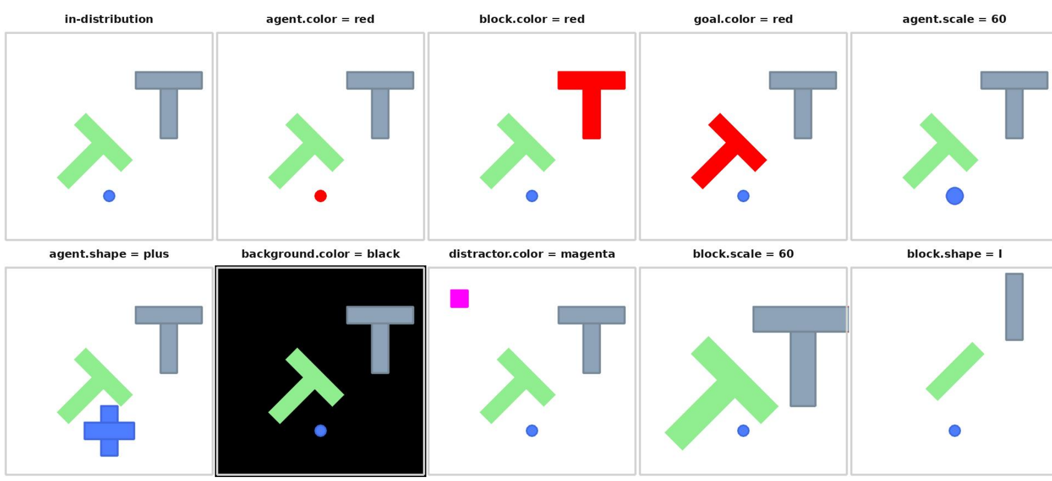


Figure 6: PushT OOD variation visualizations baseline plus one representative variation per factor. All tiles share the same fixed initial layout. Full value list is available in Table 4.

C.3 OGBench-Cube Variation Suite

Figure 7 shows one representative variation per factor; Table 5 lists all values.

C.4 OGBench-Cube Robustness Results

Figure 8 reports zero-shot planning success under the OGBench-Cube distribution shifts, complementing the PushT results in the main text (Figure 3). SlotContrast-WM and DINO-WM behave

Table 4: PushT OOD variation values. "In-dist." is the training-distribution default.

Group	Factor	In-dist. value	OOD values
appearance	`agent.color`	blue	red, yellow, black
	`block.color`	slate gray	red, blue, black
	`goal.color`	light green	red, blue, black
	`background.color`	white	red, blue, black
	`distractor.color`	off	gray, magenta
scale	`agent.scale`	40	20, 60
	`block.scale`	40	20, 60
object shape	`agent.shape`	circle	L, square, plus
	`block.shape`	T	L, square, I

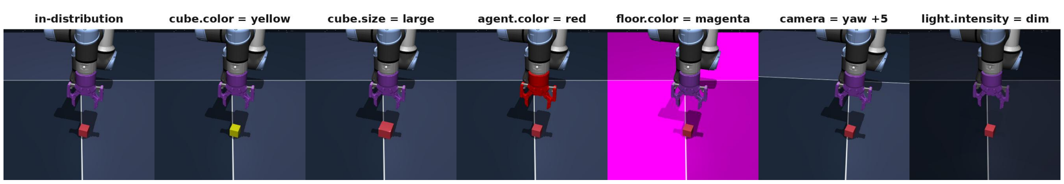


Figure 7: OGBench-Cube OOD variation visualizations: baseline and one representative variation per factor. Full value list is available in Table 5.

similarly in this environment, both staying close to their in-distribution success across all shifts. LeWM is comparatively robust to object-level variations (cube and agent color, cube size) but degrades under scene-level shifts—background color and camera angle bring it close to the 48% random-policy baseline.

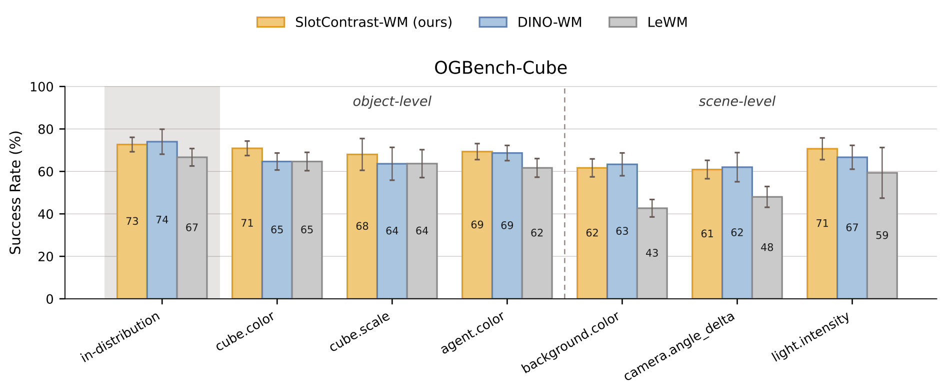


Figure 8: Zero-shot planning success rate (%) on OGBench-Cube under object-level and scene-level distribution shifts, for SlotContrast-WM, DINO-WM, and LeWM. The random-policy baseline in this environment achieves 48%.

D Extended Related Work

Unsupervised object-centric learning. Unsupervised object-centric learning (Burgess et al., 2019; Greff et al., 2019; Locatello et al., 2020) aims to decompose a scene into a small set of latent

Table 5: OGBench-Cube OOD variation values. "In-dist." is the training-distribution default.

Group	Factor	In-dist. value	OOD values
object-level	`cube.color`	crimson	red, blue, yellow
	`cube.size`	0.020 m	0.012, 0.028
	`agent.color`	purple	red, black
scene-level	`floor.color`	slate	brown, magenta
	`camera.angle_delta`	$(0°, 0°)$	yaw $\pm 5°$, pitch $\pm 5°$
	`light.intensity`	0.6	0.3, 0.95

representations, commonly referred to as slots, where each slot captures a distinct object or entity. These representations are typically learned in a self-supervised manner through reconstruction-based objectives. Slot Attention (Locatello et al., 2020) introduced an iterative attention mechanism in which slots compete to explain different regions of an image, enabling effective unsupervised object binding. Subsequent works extended this to video by adding temporal consistency and object dynamics — SAVi, SAVi++, STEVE (Kipf et al., 2022; Elsayed et al., 2022; Singh et al., 2022). More recent approaches — DINOSAUR (Seitzer et al., 2023) and the video extensions VideoSAUR, SOLV, and SlotContrast (Zadaianchuk et al., 2023; Aydemir et al., 2023; Manasyan et al., 2025) — combine slot-based architectures with pretrained vision foundation models such as DINO to improve robustness and scalability on real-world data.

Object-centric world modeling. Early OCWMs paired slot or entity encoders with relational dynamics models, including C-SWM (Kipf et al., 2020) and OP3 (Veerapaneni et al., 2019), predominantly evaluated on synthetic physics. SlotFormer (Wu et al., 2023) replaced graph-net dynamics with an autoregressive Transformer over slots, scaling to richer video datasets. More recent OCWMs target control: FOCUS (Ferraro et al., 2025) and SOLD (Mosbach et al., 2025) integrate slots into Dreamer-style model-based RL; Slot-MPC (Spieler et al., 2026) performs sampling-based MPC in slot space; Dyn-O (Wang et al., 2026) adds SAM-supervised slots for Procgen video prediction; OC-STORM (Zhang et al., 2025) pairs SAM-derived object features with a transformer world model for visually complex domains; and Feng et al. (2025) factorize representations hierarchically for policy learning.